

As $|Z| > 1.96$, the null hypothesis H_0 has to be rejected at 5% level of significance and we conclude that the die is biased.

Under large sample tests, we will see four important tests to test the significance.

1. Testing of significance for single proportion.
2. Testing of significance for difference of proportions.
3. Testing of significance for single mean.
4. Testing of significance for difference of means.

9.16 TEST OF SIGNIFICANCE OF A SINGLE MEAN - LARGE SAMPLES

Let a random sample of size n ($n \geq 30$) has the sample mean $\bar{x}$, and μ be the population mean. Also the population mean μ has a specified value μ_0 .

Working Rule :

1. **The Null Hypothesis** is $H_0 : \bar{x} = \mu$ i.e., "there is no significance difference between the sample mean and population mean" or the sample has been drawn from the parent population.
2. **The Alternative Hypothesis** is

$$(i) H_1 : \bar{x} \neq \mu \quad (\mu \neq \mu_0)$$

$$\text{or } (ii) H_1 : \bar{x} > \mu \quad (\mu > \mu_0)$$

$$\text{or } (iii) H_1 : \bar{x} < \mu \quad (\mu < \mu_0)$$

Since n is large, the sampling distribution of $\bar{x}$ is approximately normal.

3. **Level of Significance.** Set the level of significance α .
4. **The Test Statistic :**

We have the following two cases.

Case I : When the standard deviation σ of population is known.

In this case, standard Error of Mean, S. E. ($\bar{x}$) = $\frac{\sigma}{\sqrt{n}}$, where n = sample size,

σ = s.d. of population.

$\therefore$ The test statistic is

$$z = \frac{\bar{x} - \mu}{\text{S.E.}(\bar{x})} = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}, \text{ where } \mu \text{ is the sample mean.}$$

Case II : When the standard deviation σ of population is not known.

In this case, we take s , the s.d. of sample to compute the S. E. of mean

$$\therefore \text{S.E.}(\bar{x}) = \frac{s}{\sqrt{n}}$$

Hence the test statistic is $z = \frac{\bar{x} - \mu}{\text{S.E.}(\bar{x})} = \frac{\bar{x} - \mu}{s / \sqrt{n}}$

5. Find the critical value z_α of z at the level of significance α from the normal table.

6. **Decision :**

(a) If $|z| < z_\alpha$ we accept the Null Hypothesis H_0 .

(b) If $|z| > z_\alpha$, we reject the Null Hypothesis H_0 .

The rejection rule for $H_0 : \bar{x} = \mu$ (or $\mu = \mu_0$) is given below :

Table : Critical Values of z			
Level of significance α	1%	5%	10%
Critical region for $\mu \neq \mu_0$	$ z > 2.58$	$ z > 1.96$	$ z > 1.645$
Critical region for $\mu > \mu_0$	$z > 2.33$	$z > 1.645$	$z > 1.28$
Critical region for $\mu < \mu_0$	$z < -2.33$	$z < -1.645$	$z < -1.28$

Note: 1. We reject Null Hypothesis H_0 when $|z| > 3$ without mentioning any level of significance.

The test statistic is, $z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$ where σ is the S.D of the population

If the population S.D is not known, then use the statistic

$$z = \frac{\bar{x} - \mu}{s / \sqrt{n}} \text{ where } s \text{ is the sample S.D.}$$

2. The values $\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}}$ are called 95% fiducial limits or confidence limits for the mean of the population corresponding to the given sample.

Similarly, $\bar{x} \pm 2.58 \frac{\sigma}{\sqrt{n}}$ (or) $[\bar{x} - 2.58 (\text{S.E. of } \bar{x}), \bar{x} + 2.58 (\text{S.E. of } \bar{x})]$ are called

99% confidence limits and $\bar{x} \pm 2.33 \frac{\sigma}{\sqrt{n}}$ are called 98% confidence limits.

SOLVED EXAMPLES

Example 1 : According to the norms established for a mechanical aptitude test, persons who are 18 years old have an average height of 73.2 with a standard deviation of 8.6. If 4 randomly selected persons of that age averaged 76.7, test the hypothesis $\mu = 73.2$ against the alternative hypothesis $\mu > 73.2$ at the 0.01 level of significance. [JNTU 2004, 2005S (Set No. 1)]

Solution : Given $n = 4$, $\mu = 73.2$, $\bar{x}$ = mean of the sample = 76.7
and σ = S.D of population = 8.6

1. **Null Hypothesis** $H_0 : \mu = 73.2$
2. **Alternative Hypothesis** $H_1 : \mu > 73.2$ (Right-tailed test)
3. **Level of significance** : $\alpha = 99\%$ (or probability is 0.01)
4. **The test statistic** is $z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{76.7 - 73.2}{\frac{8.6}{\sqrt{4}}} = \frac{3.5}{4.3} = 0.814$

Tabulated value of z at 99% level of significance is 2.33.

Hence calculated $z <$ tabulated z

$\therefore$ The null hypothesis H_0 is accepted. That is, $\bar{x}$ and μ do not differ significantly.

Example 1 : A sample of 64 students have a mean weight of 70 kgs. Can this be regarded as a sample from a population with mean weight 56 kgs and standard deviation 25 kgs with level of significance 0.05 ? [JNTU 06, (A) Nov. 10 (Set No. 2), Dec. 2015]

Solution :

Given $\bar{x}$ = Mean of the sample = 70 kgs

μ = Mean of the population = 56 kgs

σ = S.D of population = 25 kgs

and n = Sample size = 64

1. **Null Hypothesis** H_0 : A sample of 64 students with mean weight of 70 kgs can be regarded as a sample from a population with mean weight 56 kgs and standard deviation 25 kgs.
2. **Alternative Hypothesis** H_1 : Sample cannot be regarded as one coming from the population.
3. **Level of significance** : $\alpha = 0.05$ (Given)
4. **The test statistic** is $z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{70 - 56}{\left(\frac{25}{\sqrt{64}}\right)} = 4.48$
5. The null hypothesis H_0 is rejected, since $|Z| > 1.645$

Note : The null hypothesis can be rejected even at 1% level of significance.

Example 1 : An oceanographer wants to check whether the depth of the ocean in a certain region is 57.4 fathoms, as had previously been recorded. What can he conclude at the 0.05 level of significance, if readings taken at 40 random locations in the given region yielded a mean of 59.1 fathoms with a standard deviation of 5.2 fathoms.

[JNTU 2003, 2003 S, 2004]

Solution : Given $n = 40$, $\bar{x} = 59.1$ and $\sigma = 5.2$

1. Null Hypothesis $H_0 : \mu = 57.4$
2. Alternative hypothesis $H_1 : \mu \neq 57.4$
3. Level of significance : $\alpha = 0.05$
4. The test statistic is $Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{59.1 - 57.4}{5.2/\sqrt{40}} = 2.067$

Tabulated value of Z at 5% level of significance is 1.96

Hence calculated $Z >$ tabulated Z .

$\therefore$ The null hypothesis H_0 is rejected.

Example 4 : In a random sample of 60 workers, the average time taken by them to get to work is 33.8 minutes with a standard deviation of 6.1 minutes. Can we reject the null hypothesis $\mu = 32.6$ minutes in favour of alternative null hypothesis $\mu > 32.6$ at $\alpha = 0.025$ level of significance. [JNTU 2005 (Set No. 1)]

Solution : Given $n = 60$, $\bar{x} = 33.8$, $\mu = 32.6$ and $\sigma = 6.1$

1. Null Hypothesis $H_0 : \mu = 32.6$
2. Alternative Hypothesis $H_1 : \mu > 32.6$
3. Level of significance : $\alpha = 0.025$
4. The test statistic is $Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{33.8 - 32.6}{6.1/\sqrt{60}} = \frac{1.2}{0.7875} = 1.5238$

Tabulated value of Z at 0.025 level of significance is 2.58.

Hence calculated $Z <$ tabulated Z

$\therefore$ The null hypothesis H_0 is accepted.

Example 5 : A sample of 900 members has a mean of 3.4 cms and S.D 2.61 cms. Is this sample has been taken from a large population of mean 3.25 cm and S.D 2.61 cms. If the population is normal and its mean is unknown find the 95% fiducial limits of true mean.

[JNTU (H) May 2011 (Set No. 1), (K) Nov. 2011 (Set No. 1)]

Solution :

Given $n = 900$, $\mu = 3.25$
 $\bar{x} = 3.4$ cm, $\sigma = 2.61$
 and $S = 2.61$

1. Null Hypothesis H_0 : Assume that the sample has been drawn from the population with mean $\mu = 3.25$.
2. Alternative Hypothesis $H_1 : \mu \neq 3.25$.

3. The test statistic is, $Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{3.4 - 3.25}{2.61/\sqrt{900}} = 1.724$

i.e., $Z = 1.724 < 1.96$

$\therefore$ We accept the null hypothesis H_0 .

i.e., The sample has been drawn from the population with mean $\mu = 3.25$.

95% confidence limits are given by

$$\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{n}} = 3.4 \pm 1.96 \times \frac{2.61}{\sqrt{900}} = 3.4 \pm 0.1705$$

i.e., 3.57 and 3.2295

Example 1 : A sample of 400 items is taken from a population whose standard deviation is 10. The mean of the sample is 40. Test whether the sample has come from a population with mean 38. Also calculate 95% confidence interval for the population.

[JNTU 2005, (H) Dec. 2011 (Set No. 1), (K) May 2013 (Set No. 4)]

Solution : Given $n = 400$, $\bar{x} = 40$, $\mu = 38$ and $\sigma = 10$

1. Null Hypothesis $H_0 : \mu = 38$
2. Alternative Hypothesis $H_1 : \mu \neq 38$
3. Level of significance : $\alpha = 0.05$

4. The test statistic is, $Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{40 - 38}{10/\sqrt{400}} = 4$ *i.e.*, $Z = 4 > 1.96$

$\therefore$ We reject the null hypothesis H_0 .

i.e., The sample is not from the population whose mean is 38.

95% confidence interval is $\left(\bar{x} - 1.96 \frac{\sigma}{\sqrt{n}}, \bar{x} + 1.96 \frac{\sigma}{\sqrt{n}} \right)$

i.e., $\left(40 - \frac{1.96(10)}{\sqrt{400}}, 40 + \frac{1.96(10)}{\sqrt{400}} \right)$

or $\left(40 - \frac{1.96}{20}, 40 + \frac{1.96}{20} \right)$ or $(40 - 0.98, 40 + 0.98)$

i.e., (39.02, 40.98)

Example 1 : (i) An ambulance service claims that it takes on the average less than 10 minutes to reach its destination in emergency calls. A sample of 36 calls has a mean of 11 minutes and the variance of 16 minutes. Test the claim at 0.05 level significance.

[JNTU 2005, (H) May 2012 (Set No. 4)]

(ii) An ambulance service claims that it takes on the average 8.9 minutes to reach its destination in emergency calls. To check this claim, the agency which licenses ambulance services has them timed on 50 emergency calls, getting a mean of 9.3 minutes with a standard deviation of 1.6 minutes. What can they conclude at the level of significance of 0.05 ?

[JNTU (A) June 2018]

Solution : (i) Given $n = 36$, $\bar{x} = 11$, $\mu = 10$ and $\sigma = \sqrt{16} = 4$

1. Null Hypothesis $H_0 : \mu = 10$

2. Alternative Hypothesis $H_1 : \mu < 10$

3. Level of significance : $\alpha = 0.05$

4. The test statistic is, $Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{11 - 10}{4/\sqrt{36}} = \frac{6}{4} = 1.5$

Tabulated value of Z at 5% level of significance is 1.645.

Hence calculated $Z < \text{tabulated } Z$

$\therefore$ We accept the null hypothesis H_0 .

(ii) This is left as an exercise to the reader.

Example 8 : It is claimed that a random sample of 49 tyres has a mean life of 15200 km. This sample was drawn from a population whose mean is 15150 kms and a standard deviation of 1200 km. Test the significance at 0.05 level.

[JNTU 2005, 2006S, (K) May 2013 (Set No. 2)]

Solution : Given $n = 49$, $\bar{x} = 15200$, $\mu = 15150$ and $\sigma = 1200$

1. Null Hypothesis $H_0 : \mu = 15150$

2. Alternative Hypothesis $H_1 : \mu \neq 15150$

3. Level of significance : $\alpha = 0.05$

4. Critical region : Accept the null hypothesis if $-1.96 < Z < 1.96$

5. The test statistic is, $Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{15200 - 15150}{1200/\sqrt{49}} = 0.2917$

Since $|Z| < 1.96$ therefore, we accept the null hypothesis.

Example 9 : An insurance agent has claimed that the average age of policy holders who issue through him is less than the average for all agents which is 30.5 years. A random sample of 100 policy holders who had issued through him gave the following age distribution.

Age	16-20	21-25	26-30	31-35	36-40
No. of persons	12	22	20	30	16

Calculate the Arithmetic mean and Standard deviation of this distribution and use these values to test his claim at 5% level of significance.

Solution : Take $A = 28$, $d_i = x_i - A$

$$\therefore \text{A.M.} = \bar{x} = A + \frac{h \sum f_i d_i}{N} = 28 + \frac{5 \times 16}{100} = 28.8$$

$$\begin{aligned} \text{S.D. : } S &= h \sqrt{\frac{\sum f d^2}{N} - \left(\frac{\sum f d}{N}\right)^2} = 5 \cdot \sqrt{\frac{164}{100} - \left(\frac{16}{100}\right)^2} \quad [\because h = 5] \\ &= 6.35 \end{aligned}$$

1. **Null Hypothesis H_0 :** The sample is drawn from a population with mean μ i.e. $\bar{x}$ and μ do not differ significantly where $\mu = 30.5$ years.
2. **Alternative Hypothesis H_1 :** $\mu < 30.5$ years (left tail test)

Now, $\bar{x} = 28.8$, $S = 6.35$, $\mu = 30.5$ years and $n = 100$

3. **The test statistic is,** $Z = \frac{\bar{x} - \mu}{S/\sqrt{n}} = \frac{28.8 - 30.5}{6.35/\sqrt{100}} = -2.677$

$$\therefore |Z| = 2.68.$$

Tabulated value of Z at 5% level of significance is 1.645 (left tail test).

Here calculated $Z >$ tabulated Z .

$\therefore$ The Null hypothesis H_0 is rejected.

i.e., $\bar{x}$ and μ differ significantly.

i.e., The sample is not drawn from a population with mean $\mu = 30.5$ years

Example 10 : The mean life time of a sample of 100 light tubes produced by a company is found to be 1560 hrs with a population S.D of 90 hrs. Test the hypothesis for $\alpha = 0.05$ that the mean life time of the tubes produced by the company is 1580 hrs.

[JNTU (A) Dec. 2009 (Set No. 1)]

Solution : Given $\bar{x}$ = Mean of the sample = 1560 hrs

μ = Mean of the population = 1580 hrs

n = Sample size = 100

σ = Standard deviation = 90 hrs

1. **Null Hypothesis H_0 :** $\mu = 1580$
2. **Alternative Hypothesis H_1 :** $\mu \neq 1580$
3. **Level of significance :** $\alpha = 0.05$

$$4. \text{ The test statistic is } z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{1560 - 1580}{90/\sqrt{100}} = \frac{-20}{9}$$

$$\therefore |z| = \frac{20}{9} = 2.22$$

$z_{\alpha/2}$ for $\alpha = 0.05$ is 1.96. Since $|z| > 1.96$, the null Hypothesis H_0 is rejected.

$\therefore \mu \neq 1580$.

Note: $z_{\alpha/2}$ for $\alpha = 0.01$ is 2.58

Since $|z| < 2.58$, the null Hypothesis H_0 can be accepted at 1% level of significance.

Example 11 : If the distribution of the weights of all men travelling by air between Delhi and Mumbai has a mean of 163 pounds and a standard deviation of 18 pounds. What is the probability that the consigned gross weight of 36 men travelling between these two cities is more than 6000 pounds. [JNTU (A) Dec. 2009 (Set No. 2)]

Solution : Given μ = The mean of the population = 163 pounds

σ = The standard deviation of the population = 18 pounds

Gross weight of 36 men = 6000 pounds

$\therefore \bar{x}$ = Average weight of a man = $\frac{6000}{36} = 166.7$ pounds

Also we have n = sample size = 36

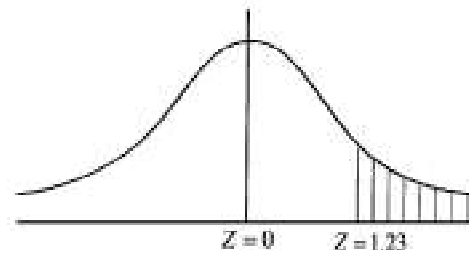
$\therefore z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{166.7 - 163}{18 / \sqrt{36}} = \frac{3.7}{3} = 1.23$

Hence $P(\bar{x} \geq 166.7) = P(Z \geq 1.23)$

$$= 0.5 - P(Z < 1.23)$$

$$= 0.5 - 0.3907 \text{ (from normal tables)}$$

$$= 0.1093$$



Thus the probability that the consigned gross weight of 36 men travelling between these two cities is more than 6000 pounds is 0.1093.

Example 11 : The length of life X of certain computers is approximately normally distributed with mean 800 hours and standard deviation 40 hours. If a random sample of 30 computers has an average life of 788 hours, test the null hypothesis that $\mu = 800$ hours against the alternative that $\mu \neq 800$ hours at (i) 0.5% (ii) 1% (iii) 4% (iv) 5% (v) 10% (vi) 15% level. [JNTU (K) May 2010 (Set No. 1)]

Solution : The Null Hypothesis is $H_0 : \mu = 800$ hours.

The Alternative Hypothesis is $H_1 : \mu \neq 800$ hours.

Since H_1 is two-tailed (i.e. two sided), we are to use two-tailed test.

Let us assume that H_0 is true.

The test statistic is $z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$

Here $\bar{x}$ = sample mean = 788 hours

$n = \text{sample size} = 30$

$\sigma = \text{standard deviation} = 40$

$$\therefore z = \frac{788 - 800}{40 / \sqrt{30}} = -1.643$$

and $|z| = 1.643$

- (i) Since $|z| < 2.81$, we accept the Null Hypothesis H_0 at 0.5% level of significance.
- (ii) Since $|z| < 2.58$, we accept the Null Hypothesis H_0 at 1% level of significance.
- (iii) Since, $|z| < 2.06$, we accept H_0 at 4% level of significance.
- (iv) Since, $|z| < 1.96$, we accept H_0 at 5% level of significance.
- (v) Since, $|z| < 1.645$, we accept H_0 at 10% level of significance.
- (vi) Since, $|z| > 1.44$, we reject H_0 at 15% level of significance.

Example 13 : In 64 randomly selected hours of production, the mean and the standard deviation of the number of acceptance pieces produced by an automatic stamping machine are $\bar{x} = 1.038$ and $\sigma = .146$

At the .05 level of significance does this enable us to reject the null hypothesis $\mu = 1.000$ against the alternative hypothesis $\mu > 1.000$? [JNTU (H) Nov, 2010 (Set No. 4)]

Solution : Let the Null Hypothesis be $H_0 : \mu = 1.000$

Then the Alternative Hypothesis is $H_1 : \mu > 1.000$

Here $\bar{x} = \text{Mean of the sample} = 1.038$

$\mu = \text{Mean of the population} = 1.000$

$\sigma = \text{S.D. of the population} = 0.146$

and $n = \text{Sample size} = 64$

$$\begin{aligned} \therefore \text{The test statistic is } z &= \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} \\ &= \frac{1.038 - 1.000}{0.146 / \sqrt{64}} = \frac{0.038}{0.146 / 8} = 2.082 \end{aligned}$$

Thus we see that $z = 2.082 > 1.645$

Hence, we reject the Null Hypothesis H_0 at 5% level of significance and conclude that the mean of the population $\mu > 1.000$

Example 14 : A trucking rm suspects the claim that average life of certain tyres is atleast 28,000 miles. To check the claim the rm puts 40 of these tyres on its trucks and gets a mean life time of 27463 miles with a standard deviation of 1348 miles. Can the claim be true? [JNTU (H) Apr. 2012 (Set No. 2)]

Solution : The Null Hypothesis is $H_0 : \mu = 28,000$ miles

The Alternative Hypothesis is $H_1 : \mu \neq 28,000$

Since H_1 is two - tailed (*i.e.*, two - sided), we are to use two - tail test.

Let us assume that H_0 is true.

The test statistic is $z = \frac{\bar{x} - \mu}{\text{S.E. of } \bar{x}} = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$

Here $\bar{x}$ = sample mean = 27463 miles, $n = 40$ and $\sigma = s = 1348$ miles

[$\because$ population S. D. σ is not known]

$$\therefore z = \frac{27463 - 28000}{1348 / \sqrt{40}} = \frac{-537(\sqrt{40})}{1348} = -2.52$$

$$\text{and } |z| = |-2.52| = 2.52 > 1.96$$

Hence, the Null Hypothesis H_0 is rejected at 5% level of significance and we conclude that the mean life of tyres, cannot be taken as 28000 miles.

9.17 TEST FOR EQUALITY OF TWO MEANS - LARGE SAMPLES

(TEST OF SIGNIFICANCE FOR DIFFERENCE OF MEANS OF TWO LARGE SAMPLES)

Let $\bar{x}_1$ and $\bar{x}_2$ be the sample means of two independent large random samples sizes n_1 and n_2 drawn from two populations having means μ_1 and μ_2 and standard deviations σ_1 and σ_2 . To test whether the two population means are equal.

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$

Then the Alternative Hypothesis is $H_1 : \mu_1 \neq \mu_2$

S. E. of $(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$, where σ_1 and σ_2 are the S. D. of the two populations

To test whether there is any significant difference between $\bar{x}_1$ and $\bar{x}_2$, we have to use the statistic

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - \delta}{\text{S.E. of } (\bar{x}_1 - \bar{x}_2)} = \frac{(\bar{x}_1 - \bar{x}_2) - \delta}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

where $\delta = \mu_1 - \mu_2$ (= given constant.)

If $\delta = 0$, the two populations have the same means.

If $\delta \neq 0$ the two populations are different.

Under $H_0: \mu_1 = \mu_2$, the test statistic becomes
$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

is approximately normally distributed with mean 0 and S. D. 1.

Note : If the samples have been drawn from the population with common S. D. σ , then $\sigma_1^2 = \sigma_2^2 = \sigma^2$.

$$\text{Hence } z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}}} = \frac{\bar{x}_1 - \bar{x}_2}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

is normally distributed with mean zero and standard deviation one.

If σ is not known we can use an estimate of σ^2 given by $\sigma^2 = \frac{n_1 S_1^2 + n_2 S_2^2}{n_1 + n_2}$

Rejection Rule for $H_0: \mu_1 = \mu_2$

- (i) If $|z| > 1.96$, reject H_0 at 5% level of significance.
- (ii) If $|z| > 2.58$, reject H_0 at 1% level of significance.
- (iii) If $|z| > 1.645$, reject H_0 at 10% level of significance.
- (iv) If $z > 3$ then either the samples have not been drawn from the same population or the sampling is not simple.

Otherwise accept H_0 .

Note : If the two samples are drawn from two populations with unknown Standard deviations σ_1^2 and σ_2^2 , then σ_1^2 and σ_2^2 can be replaced by sample variances S_1^2 and S_2^2 provided both the samples n_1 and n_2 are large.

In this case, the test statistic is
$$z = \frac{(\bar{x}_1 - \bar{x}_2) - \delta}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

Write the formula for testing the hypothesis concerning "Two Means".

[JNTU Nov. 2008 (Set No.1)]

Solution : Let $\bar{x}_1$ be the mean of a random sample of size n_1 drawn from a population with mean μ_1 and variance σ_1^2 . Let $\bar{x}_2$ be the mean of an independent random sample of size n_2 drawn from another population with mean μ_2 and variance σ_2^2 . To test the hypothesis for difference of means, the statistic is given by

$$z = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

If $|z| < 1.96$, it is accepted that the samples have come from two populations with the same mean, at 5% level of significance. Otherwise, at this level of significance, we claim that the difference in means is significant.

SOLVED EXAMPLES

Example 1 : The means of two large samples of sizes 1000 and 2000 members are 67.5 inches and 68.0 inches respectively. Can the samples be regarded as drawn from the same population of S.D 2.5 inches. [JNTU (A) Nov. 2010 (Set No. 1)]

Solution : Let μ_1 and μ_2 be the means of the two populations.

Given $n_1 = 1000$, $n_2 = 2000$ and $\bar{x}_1 = 67.5$ inches, $\bar{x}_2 = 68$ inches

Population S.D, $\sigma = 2.5$ inches

1. **Null Hypothesis H_0 :** The samples have been drawn from the same population of S.D 2.5 inches

i.e., $\mu_1 = \mu_2$ and $\sigma = 2.5$ inches

2. **Alternative Hypothesis H_1 :** $\mu_1 \neq \mu_2$

3. **The test statistic is, $z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}}} = \frac{67.5 - 68}{\sqrt{(2.5)^2 \left(\frac{1}{1000} + \frac{1}{2000} \right)}}$**

$$\Rightarrow z = \frac{-0.5}{0.0968} = -5.16$$

$\therefore |z| = 5.16 > 1.96$ i.e., the calculated value of $z >$ the table value of z .

Hence the null hypothesis H_0 is rejected at 5% level of significance and we conclude that the samples are not drawn from the same population of S.D. 2.5 inches.

Example 1 : The mean yield of wheat from a district A was 210 pounds with S.D. 10 pounds per acre from a sample of 100 plots. In another district the mean yield was 220 pounds with S.D. 12 pounds from a sample of 150 plots. Assuming that the S.D of yield in the entire state was 11 pounds, test whether there is any significant difference between the mean yield of crops in the two districts.

Solution : Let μ_1 and μ_2 be the means of the two populations.

Given $\bar{x}_1 = 210$, $\bar{x}_2 = 200$, and $n_1 = 100$, $n_2 = 150$,

Population S.D., $\sigma = 11$

1. **Null Hypothesis H_0 :** $\mu_1 = \mu_2$ i.e. there is no difference between μ_1 and μ_2

2. **Alternative Hypothesis H_1 :** $\mu_1 \neq \mu_2$

3. **The test statistic is $z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}}} = \frac{210 - 200}{\sqrt{11^2 \left(\frac{1}{100} + \frac{1}{150} \right)}} = 7.04178$**

$\therefore |z| = 7.041 > 1.96$ i.e. the calculated value of $z >$ the table value of z .

Hence we reject the Null Hypothesis H_0 at 5% level of significance and conclude that there is a significant difference between the mean yield of crops in the two districts.

Example 1 : In a survey of buying habits, 400 women shoppers are chosen at random in super market 'A' located in a certain section of the city. Their average weekly food expenditure is Rs. 250 with a S.D. of Rs. 40. For 400 women shoppers chosen at random in super market 'B' in another section of the city, the average weekly food expenditure is Rs. 220 with a S.D. of Rs. 55. Test at 1% level of significance whether the average weekly food expenditure of the two populations of shoppers are equal.

Solution : Let μ_1 and μ_2 be the means of the two populations.

Given $n_1 = 400$, $\bar{x}_1 = \text{Rs. } 250$, $S_1 = \text{Rs. } 40$

and $n_2 = 400$, $\bar{x}_2 = \text{Rs. } 220$, $S_2 = \text{Rs. } 55$

1. **Null Hypothesis H_0 :** Assume that the average weekly food expenditure of the two populations of shoppers are equal i.e., $H_0 : \mu_1 = \mu_2$
2. **Alternative Hypothesis H_1 :** $\mu_1 \neq \mu_2$

$$3. \text{ Test statistic is, } z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} = \frac{250 - 220}{\sqrt{\frac{(40)^2}{400} + \frac{(55)^2}{400}}} = \frac{30}{3.4} = 8.82$$

i.e. $z = 8.82 > 2.58$

Hence we reject the Null Hypothesis H_0 at 1% level of significance and conclude that the average weekly food expenditure of the two populations of shoppers are not equal.

Example 4 : Samples of students were drawn from two universities and from their weights in kilograms, mean and standard deviations are calculated and shown below. Make a large sample test to test the significance of the difference between the means.

	Mean	S.D	Size of the sample
University A	55	10	400
University B	57	15	100

[JNTU 2005 (Set No. 3)]

Solution :

Given $\bar{x}_1 = 55$, $\bar{x}_2 = 57$, $n_1 = 400$, $n_2 = 100$, $S_1 = 10$ and $S_2 = 15$

1. **Null Hypothesis H_0 :** $\bar{x}_1 = \bar{x}_2$ i.e., there is no difference
2. **Alternative Hypothesis H_1 :** $\bar{x}_1 \neq \bar{x}_2$
3. **Level of significance :** $\alpha = 0.05$ (assumed)
4. **Critical region :** Accept H_0 if $-1.96 < Z < 1.96$

$$5. \text{ The test statistic is } Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} = \frac{55 - 57}{\sqrt{\frac{100}{400} + \frac{225}{100}}} = \frac{-2}{\sqrt{\frac{1}{4} + \frac{9}{4}}} = -1.26$$

$$\therefore |Z| = 1.26 < 1.96$$

Hence, we accept the Null Hypothesis H_0 at 5% level of significance and conclude that there is no significant difference between the means.

Example 1 : The research investigator is interested in studying whether there is a significant difference in the salaries of MBA grades in two metropolitan cities. A random sample of size 100 from Mumbai yields on average income of Rs. 20,150. Another random sample of 60 from Chennai results in an average income of Rs. 20,250. If the variances of both the populations are given as $\sigma_1^2 = \text{Rs. } 40,000$ and $\sigma_2^2 = \text{Rs. } 32,400$ respectively.

[JNTU Nov. 2008 (Set No. 1)]

Solution : Given $n_1 = 100$, $\bar{x}_1 = 20,150$ and $n_2 = 60$, $\bar{x}_2 = 20250$

and $\sigma_1^2 = 40,000$, $\sigma_2^2 = 32400$

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$. That is, the difference of means is significant.

Then the Alternative Hypothesis is $H_1 : \mu_1 \neq \mu_2$

$$\begin{aligned} \text{The test statistic is } z &= \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{20150 - 20250}{\sqrt{\frac{40000}{100} + \frac{32400}{60}}} \\ &= \frac{100}{\sqrt{400 + 540}} = \frac{100}{30.66} = 3.26 \end{aligned}$$

Since $Z = 3.26 > 1.96 = Z_{0.05}$, we reject the Null Hypothesis at 5% level of significance and conclude that there is a significant difference in the salaries of MBA grades in two cities.

Example 1 : A researcher wants to know the intelligence of students in a school. He selected two groups of students. In the first group there 150 students having mean IQ of 75 with a S.D. of 15 in the second group there are 250 students having mean IQ of 70 with S.D. of 20. Is there a significant difference between the means of two groups ?

[JNTU Nov. 2008 (Set No. 4)]

Solution : Given $n_1 = 150$, $\bar{x}_1 = 75$, $\sigma_1 = 15$

and $n_2 = 250$, $\bar{x}_2 = 70$, $\sigma_2 = 20$

1. Null Hypothesis H_0 : The groups have been came from the same population

$$\text{i.e., } \mu_1 = \mu_2$$

2. Alternative Hypothes is $H_1 : \mu_1 \neq \mu_2$

$$\begin{aligned} \text{3. The test statistic is, } z &= \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{75 - 70}{\sqrt{\frac{225}{150} + \frac{400}{250}}} \\ &= \frac{5}{\sqrt{\frac{9}{5} + \frac{8}{5}}} = \frac{5\sqrt{5}}{\sqrt{17}} = 2.7116 \end{aligned}$$

Tabulated value of z at 1% level of significance is 2.33

$\therefore$ Calculated $z >$ tabulated z .

Hence we reject the Null Hypothesis H_0 at 1% level of significance and conclude that the groups have not been taken from the same population.

Example 1 : Two types of new cars produced in U.S.A. are tested for petrol mileage, one sample is consisting of 42 cars averaged 15 kmpl while the other sample consisting of 80 cars averaged 11.5 kmpl with population variances as $\sigma_1^2 = 2.0$ and $\sigma_2^2 = 1.5$ respectively. Test whether there is any significance difference in the petrol consumption of these two types of cars. (use $\alpha = 0.01$). [JNTU Apr. 2009 (Set No. 1)]

Solution : Let the types of the cars be named as A and B.

Number of cars of type A = 42

Average mileage for A = $\bar{x}_1 = 15$, Variance = $\sigma_1^2 = 2.0$

Number of cars of type B = 80

Average mileage for B = $\bar{x}_2 = 11.5$

Variance = $\sigma_2^2 = 1.5$

1. Null Hypothesis $H_0 : \mu_1 = \mu_2$

2. Alternative Hypothesis is $H_1 : \mu_1 \neq \mu_2$

$$\begin{aligned} \text{3. The test statistic is } z &= \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{|15 - 11.5|}{\sqrt{\frac{2}{42} + \frac{1.5}{80}}} \\ &= \frac{3.5}{\sqrt{0.0476 + 0.01875}} = \frac{3.5}{\sqrt{0.06635}} = 13.587 \end{aligned}$$

Tabulated value of z at 1% significance level is 2.58 (two-tailed)

Since $(z)_{\text{calculated}} > 2.58$ (z table), we reject Null Hypothesis H_0 at 1% level of significance and conclude that there is a significant difference in petrol consumption.

Example 8 : A simple sample of the height of 6400 Englishmen has a mean of 67.85 inches and a S. D of 2.56 inches while a simple sample of heights of 1600 Austrains has a mean of 68.55 inches and S. D. of 2.52 inches. Do the data indicate the Austrains are on the average taller than the Englishmen? (Use α as 0.01). [JNTU (A) 2009 (Set No. 2)]

Solution : We are given

n_1 = Size of the first sample = 6400

n_2 = Size of the second sample = 1600

$\bar{x}_1$ = Mean of the first sample = 67.85

$\bar{x}_2$ = Mean of the second sample = 68.55

σ_1 = Standard deviation of the first sample = 2.56

σ_2 = Standard deviation of the second sample = 2.52

1. Null Hypothesis $H_0 : \mu_1 = \mu_2$
2. Alternative Hypothesis $H_1 : \mu_1 < \mu_2$
3. Level of significance : $\alpha = 0.05$
4. The test statistic is

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\left(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)}} = \frac{67.85 - 68.55}{\sqrt{\frac{(2.56)^2}{6400} + \frac{(2.52)^2}{1600}}} = \frac{-0.7}{\sqrt{\frac{6.5536}{6400} + \frac{6.35}{1600}}}$$

$$= \frac{-0.7}{\sqrt{0.001 + 0.004}} = \frac{-0.7}{0.0707} = -9.9$$

$$\therefore |z| = 9.9 > 1.96$$

Hence, we reject the Null Hypothesis H_0 at 5% level of significance and conclude that the Australians are taller than Englishmen.

Example 9 : The mean life of a sample of 10 electric bulbs (or motors) was found to be 1456 hours with S.D. of 423 hours. A second sample of 17 bulbs (motors) chosen from a different batch showed a mean life of 1280 hours with S.D. of 398 hours. Is there a significant difference between the means of two batches? [JNTU (K) 2009, Nov. 2012 (Set No. 1)]

Solution : It is given that

n_1 = Sample size of first batch = 10

n_2 = Sample size of second batch = 17

$\bar{x}_1$ = Mean life of first batch = 1456

$\bar{x}_2$ = Mean life of second batch = 1280

σ_1 = Standard deviation of first batch = 423

σ_2 = Standard deviation of second batch = 398

1. Null Hypothesis $H_0 : \mu_1 = \mu_2$
2. Alternative Hypothesis $H_1 : \mu_1 \neq \mu_2$
3. Level of significance : $\alpha = 0.05$

$$4. \text{ The test statistic is } z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{1456 - 1280}{\sqrt{\frac{(423)^2}{10} + \frac{(398)^2}{17}}}$$

$$= \frac{176}{\sqrt{17892.9 + 9317.88}} = \frac{176}{164.96} = 1.067$$

Since $z < z_{\alpha} = 1.96$, we accept the null hypothesis H_0 i.e., there is no difference between the mean life of electric bulbs of two batches.

Example 10 : The average marks scored by 32 boys is 72 with a S.D. of 8. While that for 36 girls is 70 with a S.D. of 6. Does this indicate that the boys perform better than girls at level of significance 0.05? [JNTU (K) May 2010 (Set No. 3)]

Solution : Let μ_1 and μ_2 be the means of the two populations

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$

Then the Alternative Hypothesis is $H_1 : \mu_1 > \mu_2$

Let us assume that H_0 is true, *i.e.*, there is no difference between μ_1 and μ_2

Since the sample sizes are large we use the test statistic

$$z = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Here $\bar{x} = 72, \bar{y} = 70, \sigma_1 = 8, \sigma_2 = 6, n_1 = 32, n_2 = 36$

$$\therefore z = \frac{72 - 70}{\sqrt{\frac{64}{32} + \frac{36}{36}}} = \frac{2}{\sqrt{2+1}} = \frac{2}{\sqrt{3}} = 1.1547 < 1.96$$

Since the computed value of z is less than the table value, we cannot reject the Null Hypothesis at 5% level and conclude that the performance of boys and girls is the same.

Example 13 : At a certain large university a sociologist speculates that male students spend considerably more money on junk food than do female students. To test her hypothesis, the sociologist randomly selects from the registrar's records the names of 200 students. Of these, 125 are men and 75 are women. The sample mean of the average amount spent on junk food per week by the men is Rs. 400 and standard deviation is 100. For the women the sample mean is Rs. 450 and the sample standard deviation is Rs.150. Test the difference between the mean at .05 level. [JNTU (H) Nov. 2010 (Set No. 2)]

Solution : Let μ_1 and μ_2 be the means of the two populations.

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$

Then the Alternative Hypothesis is $H_1 : \mu_1 \neq \mu_2$

Let us assume that H_0 is true *i.e.*, there is no difference between μ_1 and μ_2 .

We are given

$$n_1 = \text{Number of men} = 125$$

$$n_2 = \text{Number of women} = 75$$

$$\bar{x}_1 = \text{Mean of men} = 400$$

$$\bar{x}_2 = \text{Mean of women} = 450$$

σ_1 = S.D. of men = 100

σ_2 = S.D. of women = 150

Level of significance, $\alpha = 0.05$

$$\begin{aligned}\text{The test statistic is } z &= \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{400 - 450}{\sqrt{\frac{(100)^2}{125} + \frac{(150)^2}{75}}} \\ &= \frac{-50}{\sqrt{80 + 300}} = \frac{-50}{\sqrt{380}} = \frac{-50}{19.49} = -2.5654\end{aligned}$$

$\therefore |z| = |-2.5654| = 2.5654 > 1.96$ i.e., the difference is highly significant.

Hence, we reject the Null Hypothesis at 5% level of significance and conclude that the two population means are not equal.

Example 11 : A company claims that its bulbs are superior to those of its main competitor. If a study showed that a sample of 40 of its bulbs have a mean life time of 647 hrs of continuous use with a S.D of 27 hrs. While a sample of 40 bulbs made by its main competitor had a mean life time of 638 hrs of continuous use with a S.D of 31 hrs. Test the significance between the difference of two means at 5% level.

[JNTU (H) Nov. 2010 (Set No. 4)]

Solution : Let μ_1 and μ_2 be the means of the two populations.

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$

Then the Alternative Hypothesis is $H_1 : \mu_1 > \mu_2$

Since the sample sizes are large, the test statistic

$$z = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

is approximately normally distributed with mean 0 and S.D. 1.

Here $\bar{x} = 647, \bar{y} = 638, \sigma_1 = 27, \sigma_2 = 31, n_1 = n_2 = 40$

$$\begin{aligned}\therefore z &= \frac{647 - 638}{\sqrt{\frac{(27)^2}{40} + \frac{(31)^2}{40}}} \\ &= \frac{9}{\sqrt{\frac{729 + 961}{40}}} = \frac{9}{6.5} = 1.38\end{aligned}$$

i.e., $|z| = 1.38 < 1.645$

Since the computed value of z is less than the table value, we cannot reject the Null Hypothesis at 5% level and conclude that the difference between the two sample means is not significant.

Example 13 : Studying the flow of traffic at two busy intersections between 4 p.m and 6 p.m to determine the possible need for turn signals. It was found that on 40 week days there were on the average 247.3 cars approaching the first intersection from the south which made left turn, while on 30 week days there were on the average 254.1 cars approaching the first intersection from the south made left turns. The corresponding sample standard deviations are 15.2 and 12. Test the significance between the difference of two means at 5% level. [JNTU (H) Dec. 2011 (Set No.1)]

Solution : Let the average cars in two places be μ_1 and μ_2 respectively.

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$

Then the Alternative Hypothesis is $H_1 : \mu_1 \neq \mu_2$

Let us assume that H_0 is true *i.e.*, there is no significant difference between μ_1 and μ_2

Since the sample sizes are large, the test statistic is

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Here $n_1 = 40, n_2 = 30, \bar{x}_1 = 247.3, \bar{x}_2 = 254.1, \sigma_1 = s_1 = 15.2$ and $\sigma_2 = s_2 = 12$.

$$\begin{aligned} \therefore z &= \frac{247.3 - 254.1}{\sqrt{\frac{(15.2)^2}{40} + \frac{(12)^2}{30}}} = \frac{-6.8}{\sqrt{5.776 + 4.8}} \\ &= \frac{-6.8}{\sqrt{10.576}} = \frac{-6.8}{3.2521} = -2.091 \end{aligned}$$

$$\therefore |z| = 2.091 > 1.96$$

Since the computed value of z is greater than the table value, we reject the Null Hypothesis at 5% level and conclude that the two average cars are significantly different *i.e.*, they are not the same in the two busy intersections.

Example 14 : In a certain factory there are two independent processes for manufacturing the same item. The average weight in a sample of 700 items produced from one process is found to be 250 gms with a standard deviation of 30 gms while the corresponding figures in a sample of 300 items from the other process are 300 and 40. Is there significant difference between the mean at 1% level. [JNTU (H) Apr. 2012 (Set No.1)]

Solution : Let the average weight in the two independent processes be μ_1 and μ_2 respectively.

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$

Then the Alternative Hypothesis is $H_1 : \mu_1 \neq \mu_2$

Let us assume that H_0 is true

i.e., there is no significant difference between μ_1 and μ_2 .

The test statistic is
$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Here $\bar{x}_1 = 250, \bar{x}_2 = 300, \sigma_1 = 30, \sigma_2 = 40$ and $n_1 = 700, n_2 = 300$

$$\therefore z = \frac{250 - 300}{\sqrt{\frac{900}{700} + \frac{1600}{300}}} = \frac{-50}{\sqrt{9 + 16}} = -19.43$$

Thus $|z| = 19.43 > 2.58$

Since the computed value of z is greater than the table value, we reject the Null Hypothesis at 1% level and conclude that there is a significant difference between the means.

Example 14 : The mean height of 50 male students who participated in sports is 68.2 inches with a S.D of 2.5. The mean height of 50 male students who have not participated in sport is 67.2 inches with a S.D of 2.8. Test the hypothesis that the height of students who participated in sports is more than the students who have not participated in sports.

[JNTU (H) Apr. 2012 (Set No. 2)]

Solution : Let the mean height in the two cases be μ_1 and μ_2 respectively.

Let the Null Hypothesis be $H_0 : \mu_1 = \mu_2$

Then the Alternative Hypothesis is $H_1 : \mu_1 \neq \mu_2$

Let us assume that H_0 is true, i.e., there is no significant difference between μ_1 and μ_2

The test statistic is
$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Here $\bar{x}_1 = 68.2, \bar{x}_2 = 67.2, \sigma_1 = 2.5, \sigma_2 = 2.8, n_1 = 50, n_2 = 50$

$$\therefore z = \frac{68.2 - 67.2}{\sqrt{\frac{(2.5)^2}{50} + \frac{(2.8)^2}{50}}} = \frac{1}{\sqrt{6.5 + 7.84}} = 1.88$$

$$\therefore |z| = 1.88 < 1.96$$

Since the computed value of z is less than the tabulated value of z , we accept the Null Hypothesis at 5% level of significance and conclude that there is no significant difference in the heights.

Example 16 : The nicotent in milligrams of two samples of tobacco were found to be as follows. Find the standard error and confidential limits for the difference between the means at 0.05 level.

Sample A	24	27	26	23	25	
Sample B	29	30	30	31	24	36

[JNTU (H) Apr. 2012 (Set No. 4)]

Solution :

Calculations for means and s_1^2, s_2^2					
Sample A			Sample B		
x_1	$x_1 - \bar{x}_1$ $= x_1 - 25$	$(x_1 - \bar{x}_1)^2$	x_2	$x_2 - \bar{x}_2$ $= x_2 - 30$	$(x_2 - \bar{x}_2)^2$
24	-1	1	29	-1	1
27	2	4	30	0	0
26	1	1	30	0	0
23	-2	4	31	1	1
25	0	0	24	-6	36
			36	6	36
$125 = \sum x_1$		10	$180 = \sum x_2$		74

$$\bar{x}_1 = \frac{\sum x_1}{n_1} = \frac{125}{5} = 25, \quad \bar{x}_2 = \frac{\sum x_2}{n_2} = \frac{180}{6} = 30$$

$$\text{Hence variance, } s_1^2 = \frac{\sum (x_1 - \bar{x}_1)^2}{n_1 - 1} = \frac{10}{4} = 2.5$$

$$\text{and variance, } s_2^2 = \frac{\sum (x_2 - \bar{x}_2)^2}{n_2 - 1} = \frac{74}{5} = 14.8$$

$$\therefore \text{ S. E. of } (\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \sqrt{\frac{2.5}{5} + \frac{14.8}{6}} = 1.72$$

Hence the 95% confidence limits are

$$(\bar{x}_1 - \bar{x}_2) \pm 1.96 (\text{S. E. of } \bar{x}_1 - \bar{x}_2)$$

$$\text{i.e., } (25 - 30) \pm 1.96 (1.72) \text{ i.e., } -5 \pm 3.37 \text{ or } (-8.37, -1.63)$$